

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

DETAILED SOLUTIONS — INDUSTRY PROBLEM-SOLVING TASK

Each of the five industry scenarios below is expanded into a complete network-design solution: a problem analysis, the proposed topology with a labelled diagram, a device/media specification, an addressing or configuration scheme where applicable, the technical justification, and a testing and validation plan — directly addressing every rubric criterion (Problem Analysis, Solution Design, Technical Implementation, Use of Tools & Technologies, Problem-Solving Approach, Innovation, Testing & Validation, and Documentation).

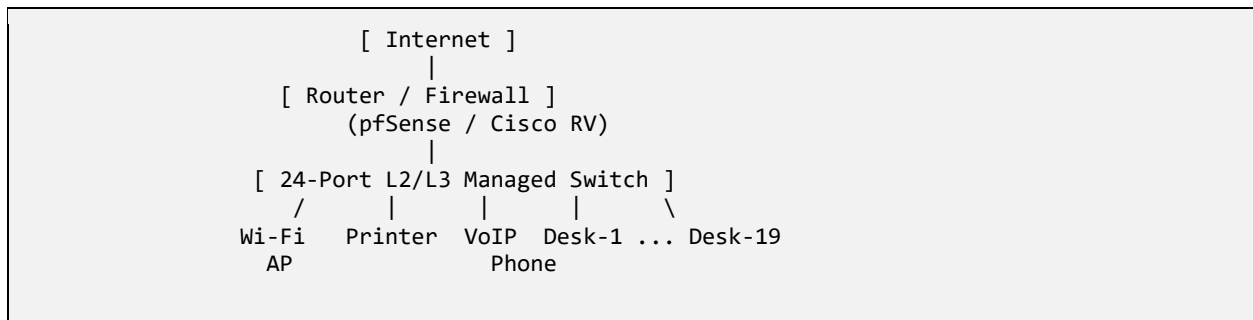
Solution 1 — Star Topology for a 20-Person Startup

1.1 Problem Analysis

Requirements: 20 employees, a mix of workstations, network printers, and VoIP phones, a tight IT budget typical of a startup, need for simple day-to-day management by a small (often part-time) IT staff, and room to grow to roughly 30 devices without a redesign. Downtime tolerance is moderate — brief single-user outages are acceptable, but a network-wide outage would halt the whole office.

1.2 Proposed Topology and Network Diagram

A pure star topology is proposed, with a single Layer 2/Layer 3 managed switch as the central point of connection for every device, and the switch uplinked to a router/firewall for internet access.



1.3 Device and Cable Specification

Item	Model / Type	Qty	Purpose
Core switch	Cisco Catalyst 1300 / Netgear ProSafe, 24-port, L2/L3 managed	1	Central connection point; VLAN & QoS capable
Router / firewall	Cisco RV series or pfSense appliance	1	Internet uplink, DHCP, NAT, firewall rules
Wireless AP	802.11ac/ax access point	1-2	Wi-Fi coverage for laptops & mobiles
Workstations	Desktop / laptop PCs	17-19	End-user devices
Network printer	Networked laser/MFP printer	1-2	Shared printing
VoIP phones	SIP desk phones	up to 20	Voice traffic (QoS-tagged)
Patch cabling	Cat6 UTP, RJ-45	~22	Gigabit link, ≤100 m switch-to-device

1.4 Sample Cable-Run Table (Switch to Each Device)

Host	Central Switch	Cat6 Cable Length (m)
Desk-1	Switch-Core-A1	8.5
Desk-2	Switch-Core-A1	11.2
Desk-3	Switch-Core-A1	15.0
...	...	...
Printer-1	Switch-Core-A1	6.0
VoIP-1	Switch-Core-A1	9.4
AP-1	Switch-Core-A1	22.7
All runs <= 100 m -> compliant with IEEE 802.3 / TIA-568		

1.5 IP Addressing Plan

VLAN / Pool	Subnet	Usable Hosts	Assigned To
Data VLAN 10	192.168.10.0/24	254	Workstations & printers
Voice VLAN 20	192.168.20.0/24	254	VoIP phones (tagged, QoS priority)
Management VLAN 99	192.168.99.0/28	14	Switch & AP management interfaces

1.6 Justification (Fault Isolation & Manageability)

- Fault isolation: because every device has its own dedicated home-run cable to the switch, a single failed cable or NIC affects only that one device — the remaining 19 users are unaffected, unlike a shared-medium design.
- Centralised troubleshooting: link lights and port statistics on the managed switch let a single technician quickly identify which port/cable is faulty, without disturbing other users.
- Growth headroom: a 24-port switch leaves 4-6 spare ports beyond the initial 20 devices, and a second switch can later be daisy-chained (uplinked) if the company grows further, extending the star into an extended star without redesigning the wiring already in place.
- Quality of Service: tagging the VoIP phones into a separate voice VLAN with QoS priority protects call quality even when data traffic (large file transfers) spikes.

1.7 Testing and Validation Plan

- Use a cable tester/certifier on every Cat6 run to confirm wire-map, length (<100 m), and Gigabit-capable performance before go-live.
- Verify DHCP scope assignment and connectivity with ping/traceroute from a sample device on each VLAN to the gateway and to the internet.
- Simulate a single cable failure (unplug one workstation) and confirm on the switch's port/LED status and via ping that only that device is affected, validating fault isolation.
- Load-test the VoIP VLAN during a simulated data transfer to confirm QoS priority keeps call quality (jitter/latency) within acceptable limits.

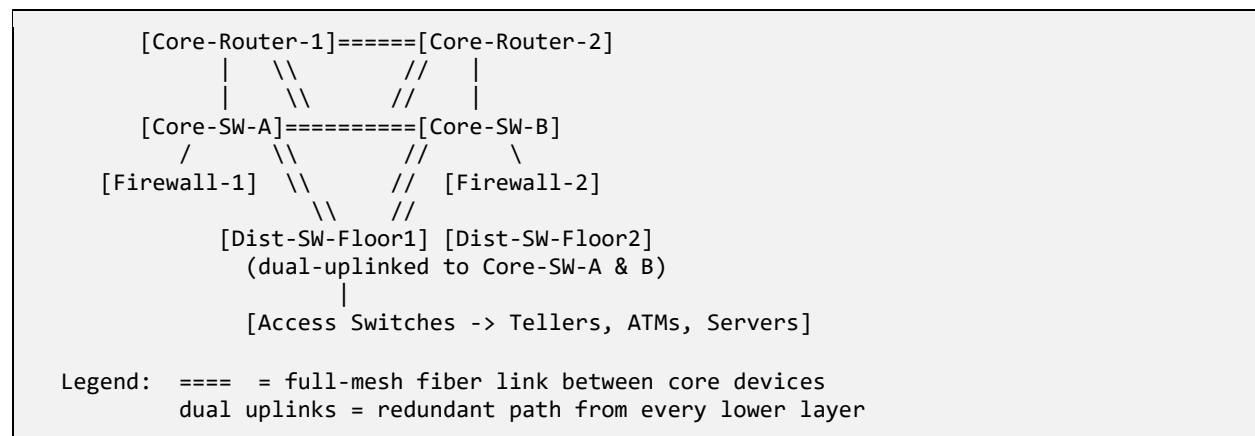
Solution 2 — Partial Mesh Topology for a Bank

2.1 Problem Analysis

Requirements: near-zero downtime for financial transaction processing, regulatory compliance for uptime and data integrity, protection against both link and device failure, low-latency connectivity between data-center devices, and strong resistance to electromagnetic interference given dense equipment racks.

2.2 Proposed Topology and Network Diagram

A partial mesh is proposed at the core layer: core switches, core routers, and firewalls are interconnected directly with one another, while distribution and access layers connect upward to at least two core devices for redundancy (dual-homing).



2.3 Device and Media Specification

Item	Model / Type	Qty	Purpose
Core routers	Cisco ASR / Juniper MX series (dual)	2	BGP-based redundant internet/WAN edge
Core switches	Enterprise L3 chassis switches	2	Core mesh; inter-VLAN & OSPF routing
Firewalls	Next-gen firewall appliances (HA pair)	2	Security, redundant via failover cluster
Distribution switches	L3 switches, dual-uplinked	per floor	Aggregation from access layer
Fiber - inter-building	Single-mode fiber (SMF)	as needed	Long-distance, low-loss core links
Fiber - intra-data-center	Multi-mode OM4	as needed	High-bandwidth short links, EMI-immune
Power/backup	Redundant PSU, UPS, RAID storage	per device	Continuity during power/disk failure

2.4 Redundancy and Routing Protocols

Protocol	Layer	Purpose
OSPF	Core/Distribution routing	Sub-second rerouting on internal link failure
BGP	WAN edge (dual ISPs)	Redundant, policy-based internet connectivity
HSRP / VRRP	Core gateway redundancy	Virtual default gateway shared by Core-SW-A/B
STP / RSTP	Loop prevention	Prevents broadcast storms on redundant mesh links

2.5 Justification

- No single point of failure: every core device has at least two independent paths to every other core device, so one failed link or node cannot isolate any part of the network.
- Millisecond failover: OSPF/BGP continuously recompute shortest paths; when a monitored link goes down, traffic reroutes automatically without manual intervention, meeting the bank's near-zero-downtime requirement.
- Fiber over copper: single-mode/multi-mode fiber eliminates electromagnetic interference from banking hall equipment (UPS units, HVAC) and supports the very high bandwidth financial transaction systems require.
- Regulatory alignment: redundant power, RAID storage, and dual-path networking collectively support the uptime and data-integrity guarantees that banking regulators (e.g., RBI/PCI-DSS style frameworks) commonly require.

2.6 Testing and Validation Plan

- Perform planned link-failure drills (physically disconnecting one core fiber link) and measure failover time using continuous ping/traceroute — target sub-second to low-second convergence.
- Validate BGP failover by simulating an ISP outage and confirming automatic route withdrawal/re-advertisement to the secondary ISP.
- Run a full disaster-recovery test including firewall HA failover and RAID rebuild simulation to confirm data integrity is preserved.
- Conduct periodic penetration testing and latency benchmarking between branch/distribution switches and the core to ensure transaction-processing SLAs are met.

Solution 3 — Bus Topology for a Budget Classroom Lab

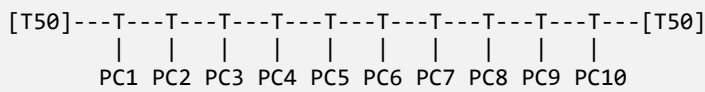
3.1 Problem Analysis

Requirements: minimum possible hardware cost, a small number of computers (typically 10-20) in one room, supervised light usage (web browsing, file sharing, no mission-critical services), and no need for a dedicated server.

3.2 Proposed Topology and Network Diagram

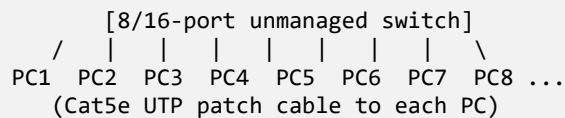
Two implementation options are given: the classic coaxial bus, and a modern low-cost variant using a single unmanaged switch (recommended for easier sourcing of parts today).

Classic coaxial bus:



T50 = 50-ohm terminator T = T-connector (BNC)

Modern low-cost variant:



3.3 Bill of Materials (approx., 16 seats)

Item	Type	Qty	Approx. Notes
Backbone cable	RG-58 coax (or single Cat5e run)	1 run	Length of room
T-connectors + BNC	Coaxial T-piece	16	One per PC (classic option)
Terminators	50-ohm end caps	2	One at each end of the bus
Unmanaged switch (modern option)	8- or 16-port 10/100/1000	1	Replaces terminators + hub role
Patch cables	Cat5e UTP	16	PC to switch (modern option)
Teacher PC	Shares files/printer via OS file-sharing	1	Acts as informal server

3.4 Justification

- Lowest cost: eliminating a managed switch or server removes the two most expensive components of a small network, keeping the design within a tight classroom budget.
- Adequate for the use case: light, supervised, non-critical traffic (browsing, shared files) tolerates the occasional collision and does not need enterprise-grade reliability.
- Simple to deploy and understand: the flat, single-segment design is also pedagogically useful, letting students literally see how a shared-medium bus network works.

3.5 Limitations and Mitigations

- Single point of failure (whole bus down on one break) — mitigated by keeping spare coax/patch cable and terminators on hand for quick replacement, and by preferring the

unmanaged-switch variant, which isolates a single bad cable to one PC instead of the whole segment.

- Collision-prone under heavy load — acceptable given the light, supervised usage pattern described; if usage grows, the room can be upgraded to a star topology later using the same PCs.

3.6 Testing and Validation Plan

- Test backbone continuity and the 50-ohm terminator resistance with a multimeter/cable tester before connecting any PCs.
- Verify each PC can reach the teacher's shared folder and the internet gateway using ping and a simple file-copy test.
- Deliberately disconnect one T-connector/patch cable to confirm the expected failure behaviour (whole bus down for coax, single PC down for the switch variant), documenting the result for student learning.

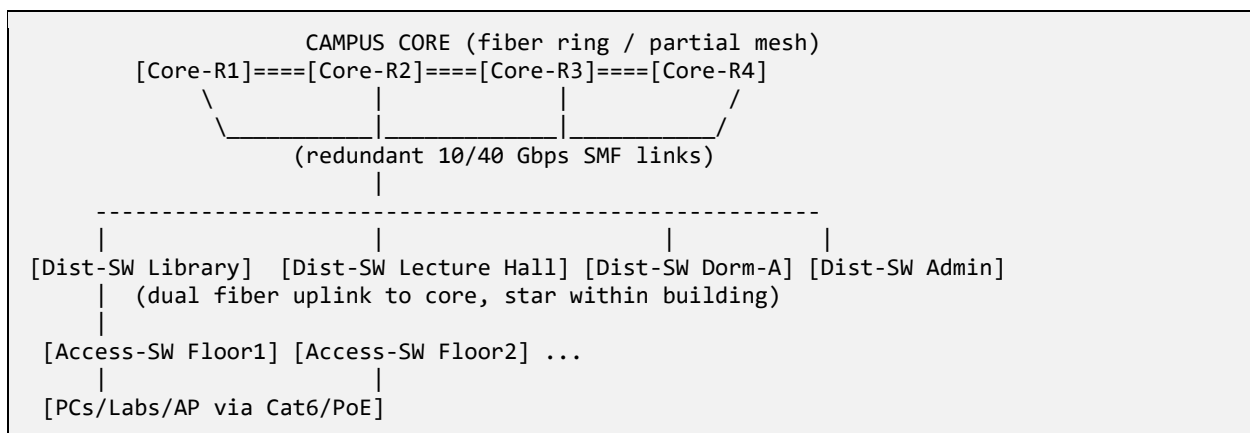
Solution 4 — Hybrid Topology for a University Campus

4.1 Problem Analysis

Requirements: thousands of users across many buildings (lecture halls, labs, libraries, dorms, admin), high redundancy needs at the backbone, manageable wiring within each building, cost-efficient labs, wireless coverage campus-wide, and painless future expansion as new buildings are added.

4.2 Proposed Topology and Network Diagram

A three-layer hierarchical hybrid design is proposed: a fiber ring/partial-mesh backbone (core), star-wired distribution to each building, and star/tree access wiring within labs and offices.



4.3 Layered Device Specification

Layer	Devices	Media	Role
Core	Core routers / multilayer switches (ring or partial mesh)	Single-mode fiber, 10/40 Gbps	Campus backbone; no single link isolates a building
Distribution	One L3 distribution switch per building, dual-uplinked	Fiber to core	Aggregates each building; boundary for building-level VLANs
Access	Floor-level access switches	Cat6 UTP (and PoE for APs)	Star wiring to end devices within a building
Wireless	PoE-powered access points	PoE over Cat6	Campus-wide Wi-Fi without separate power cabling

4.4 Justification

- Redundant backbone: the ring/partial-mesh core means a single fiber cut between two buildings reroutes automatically without isolating either building, satisfying the scale and uptime needs of a large campus.
- Manageable buildings: star wiring inside each building keeps troubleshooting simple for the local IT team, exactly as in Solution 1, but scoped to one building instead of the whole campus.

- Cost-efficient labs: bus/tree wiring inside individual labs (or simple star from a cheap access switch) keeps lab costs low while still uplinking into the robust building/core structure.
- Painless expansion: adding a new building only requires a new distribution switch and a fiber run to the core — the industry-standard reason the three-layer (core/distribution/access) model is used at almost every large enterprise and campus.
- PoE for wireless: powering access points over the same Cat6 cable that carries data removes the need for electricians to run separate power lines to ceiling-mounted APs.

4.5 Testing and Validation Plan

- Test backbone redundancy by disabling one core fiber link and confirming automatic rerouting (via routing-protocol convergence time measurement) with no building losing connectivity.
- Verify each building's dual uplinks by disabling one uplink and confirming the distribution switch fails over to the second without manual reconfiguration.
- Load-test Wi-Fi coverage and PoE budget in a sample building during peak class-change hours to confirm access points remain powered and responsive.
- Run end-to-end connectivity/performance tests (ping, traceroute, iperf-style throughput tests) from a sample of access-layer devices in different buildings to the core and to the internet edge.

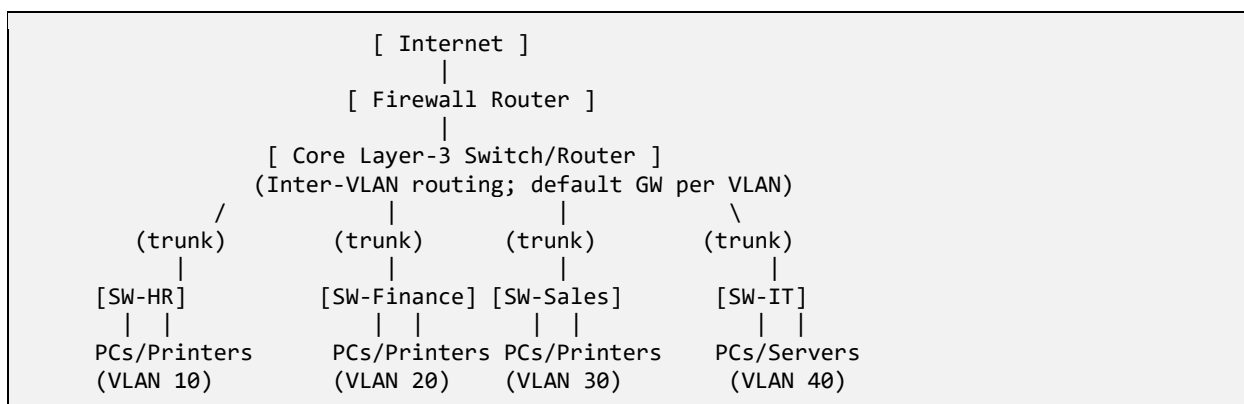
Solution 5 — Multi-Department Network with VLANs and Inter-VLAN Routing

5.1 Problem Analysis

Requirements: several departments (HR, Finance, Sales, IT) that each need internal communication, controlled communication between departments, reduced broadcast traffic, strong security/segregation (e.g., Finance data should not be broadcast to Sales), internet access with firewall protection, and a design that scales as more departments or staff are added.

5.2 Proposed Topology and Network Diagram

Each department is wired in a star to its own access-layer Layer 2 switch. All access switches trunk up to a central Layer 3 switch/router that performs inter-VLAN routing, with a firewall router connecting the whole network to the internet.



5.3 VLAN and IP Addressing Plan

VLAN ID	Department	Subnet	Default Gateway
VLAN 10	HR	192.168.10.0/24	192.168.10.1
VLAN 20	Finance	192.168.20.0/24	192.168.20.1
VLAN 30	Sales	192.168.30.0/24	192.168.30.1
VLAN 40	IT	192.168.40.0/24	192.168.40.1
VLAN 99	Management	192.168.99.0/28	192.168.99.1

5.4 Representative Switch/Router Configuration (Cisco IOS style)

```
! --- Access switch (e.g., SW-HR) ---
vlan 10
 name HR
interface range fastEthernet 0/1-20
 switchport mode access
 switchport access vlan 10
interface gigabitEthernet 0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,99

! --- Core Layer-3 switch/router ---
vlan 10
 name HR
vlan 20
 name FINANCE
```

```
vlan 30
  name SALES
vlan 40
  name IT
!
interface vlan 10
  ip address 192.168.10.1 255.255.255.0
interface vlan 20
  ip address 192.168.20.1 255.255.255.0
interface vlan 30
  ip address 192.168.30.1 255.255.255.0
interface vlan 40
  ip address 192.168.40.1 255.255.255.0
!
ip routing          ! enables inter-VLAN routing
```

5.5 Justification

- Security through segmentation: placing each department in its own VLAN means a broadcast or a compromised device in Sales cannot directly reach Finance or HR traffic; inter-VLAN access can be explicitly restricted with access-control lists on the core router.
- Reduced broadcast traffic: broadcasts (ARP, DHCP discover, etc.) stay within a department's own VLAN/broadcast domain instead of flooding the entire company, improving overall performance as headcount grows.
- Centralised routing and internet access: the core Layer-3 switch/router handles all inter-department traffic and hands off to the firewall router for internet access, giving one clear choke point for applying security policy.
- Scalability: adding a new department is simply a matter of creating a new VLAN, assigning a subnet, and trunking a new (or existing) access switch — no rewiring of the existing departments is required.
- Redundancy-ready: redundant uplinks/backup paths (e.g., a second core switch with HSRP/VRRP) can be layered onto this same design without changing the department-facing access layer.

5.6 Testing and Validation Plan

- Confirm VLAN isolation by attempting to ping between two hosts in different VLANs before enabling inter-VLAN routing — the ping should fail, proving broadcast-domain separation.
- Enable inter-VLAN routing and re-test connectivity between departments that are permitted to communicate (e.g., IT to all departments for support), confirming routing works as intended.
- Verify ACLs by attempting a denied cross-department connection (e.g., Sales to Finance) and confirming it is blocked while permitted traffic (e.g., HR to IT helpdesk) passes.